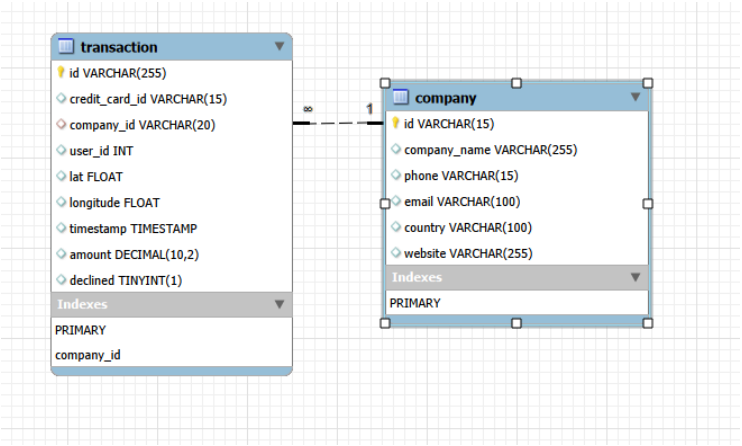
Tarea S2.01. Nociones básicas SQL

Nivel 1

- Ejercicio 1

A partir de los documentos adjuntos, importa las dos tablas. Muestra las características principales del esquema creado y explica las diferentes tablas y variables que existen. Asegúrate de incluir un diagrama que ilustre la relación entre las diferentes tablas y variables.



Tenemos dos tablas “transaction” y “company” con una relación N a 1 entre ellas ambas con un campo “id” como clave primaria.

Están vinculadas entre sí, por la clave foránea “company_id” en la tabla “transaction” que se vincula con la clave primaria “id” de la tabla “company”.

Se puede deducir una estructura en estrella, donde la tabla de hechos es “transaction” y “company” es una tabla de dimensión.

En la tabla “transaction” tenemos los registros de transacciones: identificador de transacción, tarjeta de crédito, compañía, usuario, y datos sobre la transacción, es una tabla de datos transaccionales.

En la tabla “company” tenemos los registros de empresas: identificador de empresa, nombre, teléfono, correo, país, web, es una tabla de datos maestros.

- Ejercicio 2 (Utilizando JOIN realizarás las siguientes consultas:)

Listado de los países que están generando ventas.

```
SELECT DISTINCT c.country
FROM company AS c
INNER JOIN transaction AS t ON t.company_id = c.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country
Germany
Australia
United States
New Zealand
Norway
United Kingdom

Result 18

Output

Action Output

#	Time	Action	Message	Duration /
100161	13:04:42	SELECT DISTINCT c.country FROM company AS c INNER JOIN transaction AS t ON t.company_i...	15 row(s) returned	0.015 sec

Desde cuántos países se generan las ventas.

```
SELECT COUNT(DISTINCT c.country)
FROM company AS c
INNER JOIN transaction AS t ON t.company_id = c.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(DISTINCT c.country)
15

Result 19

Output

Action Output

#	Time	Action	Message	Duration / Fetc
100164	13:27:28	SELECT DISTINCT c.country FROM company AS c INNER JOIN transaction AS t ON t.company_i...	15 row(s) returned	0.015 sec / 0.0
100165	13:29:15	SELECT COUNT(DISTINCT c.country) FROM company AS c INNER JOIN transaction AS t ON t.co...	1 row(s) returned	0.140 sec / 0.0

Identifica la compañía con la media más grande de ventas.

```
SELECT c.id, c.company_name, AVG(t.amount) AS avg_sales
FROM company AS c
INNER JOIN transaction AS t ON t.company_id = c.id
GROUP BY c.id, c.company_name
ORDER BY avg(t.amount) DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id	company_name	avg_sales
b-2222	Ac Fermentum Incorporated	284.867160

Result 21

Output

Action Output

#	Time	Action	Message	Duration / Fetc
100166	13:32:14	SELECT c.id, c.company_name, AVG(t.amount) AS avg_sales FROM company AS c INNER JOIN tr...	100 row(s) returned	0.531 sec / 0.
100167	13:32:49	SELECT c.id, c.company_name, AVG(t.amount) AS avg_sales FROM company AS c INNER JOIN tr...	1 row(s) returned	0.437 sec / 0.

- Ejercicio 3 Utilizando solo subconsultas (sin utilizar JOIN):

Muestra todas las transacciones realizadas por empresas de Alemania.

```
SELECT *
FROM transaction
WHERE company_id IN (
    SELECT id
    FROM company
    WHERE country = 'Germany'
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0
00687139-48B2-4FFA-8E73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4669	2019-03-09 19:37:49	524.84	0
0074F4DD-32F1-4827-8758-55896314623A	CcS-8081	b-2222	3500	39.7016	-8.50325	2016-12-26 23:06:57	491.90	0
00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9	CcS-6797	b-2222	2216	55.7652	-3.76245	2021-04-25 03:06:59	167.15	0
00BE09D4-6920-47D8-ABE8-325E2269829D	CcS-4983	b-2222	402	38.708	-9.12993	2019-02-27 15:25:16	141.66	0

transaction 22

Output

Action Output

#	Time	Action	Message	Duration /
100167	13:32:49	SELECT c.id, c.company_name, AVG(t.amount) AS avg_sales FROM company AS c INNER JOIN tr...	1 row(s) returned	0.437 sec /
100168	13:35:20	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE c...	1000 row(s) returned	0.000 sec /

Lista las empresas que han realizado transacciones por un amount superior a la media de todas las transacciones.

```
45 • SELECT company_name
46   FROM company
47   WHERE id IN (
48       SELECT company_id
49       FROM transaction
50       WHERE amount > ( SELECT AVG(amount) FROM transaction )
51   );
52
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name
Ac Fermentum Incorporated
Magna A Neque Industries
Fusce Corp.
Convallis In Incorporated
Ante Iaculis Nec Foundation
Donec Ltd
Sed Nunc Ltd
Amet Nulla Donec Corporation
Nascetur Ridiculus Mus Inc.

company 23

Output

Action Output

#	Time	Action	Message	Duration /
100168	13:35:20	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE c...	1000 row(s) returned	0.000 sec
100169	13:36:45	SELECT company_name FROM company WHERE id IN (SELECT company_id FROM transaction ...	100 row(s) returned	0.140 sec

Eliminarán del sistema las empresas que no tienen transacciones registradas, entrega el listado de estas empresas.

```
55 • SELECT c.id
56   FROM company c
57   WHERE NOT EXISTS (
58       SELECT 1
59       FROM transaction t
60       WHERE t.company_id = c.id
61   );
62
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id
NULL

company 24

Output

Action Output

#	Time	Action	Message	Duration / Fetc
100169	13:36:45	SELECT company_name FROM company WHERE id IN (SELECT company_id FROM transaction ...	100 row(s) returned	0.140 sec / 0.0
100170	13:37:31	SELECT c.id FROM company c WHERE NOT EXISTS (SELECT 1 FROM transaction t WH...	0 row(s) returned	0.016 sec / 0.0

Nivel 2

- Ejercicio 1

Identifica los cinco días que se generó la cantidad más grande de ingresos a la empresa por ventas. Muestra la fecha de cada transacción junto con el total de las ventas.

```
SELECT DATE(timestamp) AS date_transaction, SUM(amount) AS total_sales
FROM transaction
GROUP BY DATE(timestamp)
ORDER BY SUM(amount) DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

date_transaction	total_sales
2022-12-13	14337.44
2019-11-18	13591.32
2023-02-20	13332.59
2017-12-20	13318.43
2019-03-18	12680.95

Result 25 | Read C

Output

Action Output

#	Time	Action	Message	Duration
100170	13:37:31	SELECT c.id FROM company c WHERE NOT EXISTS (SELECT 1 FROM transaction t WH...	0 row(s) returned	0.016 sec
100171	13:41:45	SELECT DATE(timestamp) AS date_transaction, SUM(amount) AS total_sales FROM transaction G...	5 row(s) returned	0.172 sec

- Ejercicio 2

¿Cuál es la media de ventas por país? Presenta los resultados ordenados de mayor a menor medio.

```
SELECT c.country, AVG(t.amount) AS avg_sales
FROM company AS c
INNER JOIN transaction AS t ON c.id = t.company_id
GROUP BY c.country
ORDER BY AVG(t.amount) DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	avg_sales
Australia	265.190742
United States	264.977877
Belgium	261.153042
Germany	260.841391
Ireland	260.644761
Spain	260.468125
France	259.979185
New Zealand	259.586176

Result 26

Output

Action Output

#	Time	Action	Message
100171	13:41:45	SELECT DATE(timestamp) AS date_transaction, SUM(amount) AS total_sales FROM transaction G...	5 row(s) returned
100172	13:44:18	SELECT c.country, AVG(t.amount) AS avg_sales FROM company AS c INNER JOIN transaction AS ...	15 row(s) returned

- Ejercicio 3

En tu empresa, se plantea un nuevo proyecto para lanzar algunas campañas publicitarias para hacer competencia a la compañía "Non Institute". Para lo cual, te piden la lista de todas las transacciones realizadas por empresas que están situadas en el mismo país que esta compañía.

Muestra el listado aplicando JOIN y subconsultas.

```
SELECT t.*
FROM company AS c
INNER JOIN transaction AS t ON c.id = t.company_id
WHERE c.country = (
    SELECT country
    FROM company
    WHERE company_name = "Non Institute"
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
008629B4-C9A9-406C-A3D2-71FDA47BC546	CcS-7063	b-2246	2482	45.7666	4.83048	2015-07-30 12:12:42	486.44	0
00B72BA4-54A3-4B8E-B13F-2D57535AA17A	CcS-8475	b-2246	3894	55.6212	-3.7546	2017-10-26 22:08:26	414.06	0
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C	CcS-8700	b-2246	4119	55.856	-3.15783	2018-01-27 13:44:36	103.73	0
023FFCE8-E618-4938-BF56-C8DF80540ADD	CcS-7816	b-2246	3235	46.3568	1.82755	2016-12-19 11:53:45	219.28	0
02683BEB-EF91-4564-957B-D6F1662AB7C5	CcS-9471	b-2246	4890	42.1332	12.396	2017-01-10 21:09:29	326.87	0
02C2F29E-CEF2-4C1E-A594-F476E8F279C0	CcS-9082	b-2246	4501	39.4662	-0.373246	2020-05-24 01:17:29	155.72	0
02F468DC-426C-47C2-8B0A-D8B25B7A81AF	CcS-6913	b-2246	2332	52.175	19.3508	2023-03-17 16:36:27	305.35	0
0306BE3B-817B-4A49-934E-0E439291A104	CcS-5302	b-2246	721	51.9233	18.926	2021-12-02 23:06:02	339.58	0
0347BFE6-8EB5-4387-B187-0E78E8F2B8FB	CcS-7674	b-2246	3093	45.768	4.84271	2021-12-30 08:40:24	172.93	0

Result 27

Output

Action Output

#	Time	Action	Message	Duration /
100172	13:44:18	SELECT c.country, AVG(t.amount) AS avg_sales FROM company AS c INNER JOIN transaction AS ...	15 row(s) returned	0.422 sec
100173	13:45:45	SELECT t.* FROM company AS c INNER JOIN transaction AS t ON c.id = t.company_id WHERE c....	1000 row(s) returned	0.016 sec

Muestra el listado aplicando solo subconsultas.

```
SELECT *
FROM transaction
WHERE company_id IN (
    SELECT id
    FROM company
    WHERE country IN (
        SELECT country
        FROM company
        WHERE company_name = "Non Institute"
    )
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
003AD3F7-AF0F-441C-B221-6117814CA2F7	CcS-6228	b-2618	1647	55.3732	-3.39702	2021-01-14 20:39:57	266.68	0
00C26276-9500-4D80-A2BE-49A064C229FB	CcS-9064	b-2618	4483	39.4627	-0.377206	2016-03-24 08:06:04	136.60	0
00CA0053-4EA7-4F80-AE4E-0E7AB6D65CFF	CcS-8527	b-2618	3946	46.5307	2.27639	2016-09-19 22:11:41	437.09	0
00D7ECC1-F470-40CD-8DD0-8A7F709F76C7	CcS-8926	b-2618	4345	46.5688	1.93599	2020-01-02 13:10:34	345.92	0
00DC1967-4260-458B-926A-35DB317433C5	CcS-6473	b-2618	1892	52.3388	5.7368	2023-05-05 07:46:56	62.62	0
00E4DA17-35F6-4D61-99E7-B317455DB1B6	CcS-8954	b-2618	4373	42.2102	12.1252	2017-08-11 11:12:53	92.91	0
0147854E-0AEB-4A32-B26A-4F0C1B73C051	CcS-6381	b-2618	1800	51.7762	5.56833	2018-11-15 17:00:14	270.93	0
0173575D-6BE2-4450-AFED-9568E3549FBA	CcS-9255	b-2618	4674	39.0251	-7.97634	2021-11-20 11:00:18	30.14	0
0175A14D-921F-4D71-B3E8-8AE2B4074DC2	CcS-6339	b-2618	1758	41.3898	2.16622	2018-05-11 07:19:30	493.06	0
018FC5FC-6619-4D45-BBEC-445886FCDA42	CcS-7065	b-2618	2484	60.1232	18.4976	2023-06-30 09:37:11	526.48	0
01F7B6C6-F4C0-4D83-94D3-4DDDA41893C0	CcS-6388	b-2618	1807	41.7263	13.0113	2018-03-02 05:58:33	226.34	0
020614D3-07AE-47FD-BD47-2B83442C894F	CcS-7922	b-2618	3341	39.448	-8.52514	2024-08-05 10:51:44	189.21	0
023D1462-1AFC-4B71-B65C-3BA1FF496688	CcS-8011	b-2618	3430	47.0052	1.6806	2016-04-14 13:29:06	203.06	0

transaction 29

Output

Action Output

#	Time	Action	Message
100174	13:46:48	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE c...	1000 row(s) returned
100175	13:47:14	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE c...	1000 row(s) returned

Nivel 3

- Ejercicio 1

Presenta el nombre, teléfono, país, fecha y amount, de aquellas empresas que realizaron transacciones con un valor comprendido entre 350 y 400 euros y en alguna de estas fechas: 29 de abril del 2015, 20 de julio del 2018 y 13 de marzo del 2024. Ordena los resultados de mayor a menor cantidad.

```
SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS date, t.amount
FROM company AS c
INNER JOIN transaction AS t ON t.company_id = c.id
WHERE DATE(t.timestamp) IN ('2015-04-29','2018-07-20','2024-03-13')
AND t.amount BETWEEN 350 AND 400
ORDER BY t.amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	date	amount
Aliquam PC	01 45 73 52 16	Germany	2024-03-13	399.84
Auctor Mauris Vel LLP	08 09 28 74 14	United States	2018-07-20	399.51
At Pede Corp.	06 14 48 33 15	Italy	2015-04-29	390.69
Aliquam PC	01 45 73 52 16	Germany	2024-03-13	388.29
Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	2018-07-20	373.71
Fringilla LLC	08 29 15 93 57	New Zealand	2015-04-29	367.62
Pede Cum Ltd	07 62 26 48 38	Norway	2018-07-20	356.87
Auctor Mauris Vel LLP	08 09 28 74 14	United States	2024-03-13	353.75

Result 30

Output

Action Output

#	Time	Action	Message	Duration / Fetc
100175	13:47:14	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE c...	1000 row(s) returned	0.000 sec / 0.
100176	13:48:56	SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS date, t.amount FROM compa...	8 row(s) returned	0.125 sec / 0.

- Ejercicio 2

Necesitamos optimizar la asignación de los recursos y dependerá de la capacidad operativa que se requiera, por lo cual te piden la información sobre la cantidad de transacciones que realicen las empresas, pero el departamento de recursos humanos es exigente y quiere un listado de las empresas donde especifiques si tienen más de 400 transacciones o menos.

```
SELECT c.*, t.n_transactions
FROM company AS c
INNER JOIN (
SELECT
    company_id,
    CASE
        WHEN COUNT(id) >= 400 THEN '400 o més'
        ELSE 'menys de 400'
    END AS n_transactions
FROM transaction
GROUP BY company_id
) AS t ON c.id = t.company_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	company_name	phone	email	country	website	n_transactions
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site	400 o més
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9	400 o més
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars	400 o més
b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.couk	Germany	https://cnn.com/user/110	400 o més
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netflix.com/settings	400 o més
b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.couk	Norway	https://nytimes.com/user/110	400 o més
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one	400 o més
b-2250	Amet Nulla Donec Corporation	07 15 25 14 74	mattis.integer.eu@protonmail.net	Italy	https://netflix.com/sub/cars	400 o més
b-2254	Nascetur Ridiculus Mus Inc.	06 26 87 61 84	suspendisse.dui@icloud.net	United States	https://ebay.com/sub	400 o més
b-2258	Vestibulum Lorem PC	02 02 87 33 40	aenean.massa.integer@aol.net	Belgium	https://pinterest.com/sub/cars	400 o més
b-2262	Gravida Sagittis LLP	03 81 28 33 97	turpis.vitae@google.ca	Sweden	https://naver.com/site	400 o més
b-2266	Mus Aenean Eget Foundation	06 25 15 52 43	mi.duis@hotmail.net	Sweden	https://instagram.com/group/9	400 o més

Result 31

Output

Action Output

#	Time	Action	Message	Duration /
100176	13:48:56	SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS date, t.amount FROM compa...	8 row(s) returned	0.125 sec
100177	13:53:36	SELECT c.*, t.n_transactions FROM company AS c INNER JOIN (SELECT company_id, CASE W...	100 row(s) returned	0.062 sec